

Sumcheck-free hypercube inner products from table-form Kronecker encodings

Research note (companion to Kronecker-FRI)

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Abstract

We encode a multilinear polynomial by its *table*, $V_f(X) = \sum_w f(w)X^w$, instead of its coefficients. Two quantities then become single coefficients of univariate products: the evaluation $\tilde{f}(z)$, with a product-form kernel, and the hypercube inner product $\sum_w a(w)b(w)$ of two committed tables, through the reversal $V_b^*(X) = X^{N-1}V_b(1/X)$. Because a smooth domain is closed under inversion, the verifier reads V_b^* from the commitment to b as a virtual word, and the Kronecker-FRI machinery (opening polynomial, virtual word, one folding test) proves $\sum_w a(w)b(w) = S$ with no sumcheck, with soundness error $3M/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1-\delta)^\kappa$. We state precisely what remains for a sumcheck-free Hadamard (zero-)check, the degree-3 statement that SNARKs need.

Setting. Notation and parameters are those of Kronecker-FRI (KF) [1]: $n \geq 1$, $N = 2^n$, a smooth domain $L \leq \mathbb{F}^\times$ of order $M = 2^{n+R}$ with $R \geq 2$, codes $\mathcal{C}_j$, the folding test $\Pi_{\text{FT}}[\ell, \kappa]$ with error $\varepsilon_{\text{fold}}$ (KF Thm 5.24), fibre distance Δ_0^{fib} , and $0 < \delta \leq \delta^* = (1 - \rho)/2$. A bit vector $w \in \{0, 1\}^n$ is identified with the integer $\sum_k w_k 2^{k-1} \in [0, N - 1]$.

1 Table encoding and two extraction identities

Definition 1 (Table encoding, reversal). For a table $f : \{0, 1\}^n \rightarrow \mathbb{F}$, $V_f(X) = \sum_w f(w)X^w \in \mathbb{F}[X]_{<N}$ and $\text{Enc}^T(f) = \text{ev}_L(V_f)$. For $P \in \mathbb{F}[X]_{<N}$, its reversal is $P^*(X) = X^{N-1}P(1/X)$, so $[X^j]P^* = [X^{N-1-j}]P$ and $(P^*)^* = P$.

$f \mapsto V_f$ is a linear bijection onto $\mathbb{F}[X]_{<N}$, so Enc^T is a linear bijection onto $\mathcal{C}_0$ (as KF Lemma 6.3(1)); it is the Kronecker encoding of the multilinear polynomial whose coefficients are the values of f , and needs no Möbius transform. In particular $\text{Enc}^T(f) = \text{Enc}(f)$ as maps on vectors of length N : a table-form commitment is an ordinary Kronecker-FRI commitment to the vector of values, with the same Merkle tree, and all of KF's results on commitments (unique decoding, extraction, batching) apply to it unchanged.

Lemma 2 (Evaluation). For $z \in \mathbb{F}^n$ let $E_z^*(X) = \prod_{k=1}^n (z_k + (1 - z_k)X^{2^{k-1}})$. Then $\tilde{f}(z) = [X^{N-1}]V_f E_z^*$, where $\tilde{f}$ is the multilinear extension of f .

Proof. Expanding the product, $E_z(X) := \sum_w \text{eq}(w, z)X^w = \prod_k ((1 - z_k) + z_k X^{2^{k-1}})$, and $X^{N-1}E_z(1/X) = \prod_k X^{2^{k-1}}((1 - z_k) + z_k X^{-2^{k-1}}) = E_z^*(X)$ since $N - 1 = \sum_k 2^{k-1}$. Hence $[X^{N-1-w}]E_z^* = [X^w]E_z = \text{eq}(w, z)$ and $[X^{N-1}]V_f E_z^* = \sum_w f(w) \text{eq}(w, z) = \tilde{f}(z)$. $\square$

Lemma 3 (Inner product). For tables a, b : $\sum_w a(w)b(w) = [X^{N-1}]V_a V_b^*$.

Proof. $[X^{N-1}]V_a V_b^* = \sum_{i=0}^{N-1} [X^i]V_a [X^{N-1-i}]V_b^* = \sum_i [X^i]V_a [X^i]V_b = \sum_w a(w)b(w)$. $\square$

Both are instances of one fact: for $U \in \mathbb{F}[X]_{<N}$ and a public $K \in \mathbb{F}[X]_{<N}$, $[X^{N-1}]UK = \sum_i [X^i]U[X^{N-1-i}]K$. KF's split lemma (Lemma 4.10) and identity lemma (Lemma 4.12) use only $\deg U, \deg K \leq N-1$, so they hold verbatim for UK replaced by $V_a E_z^*$ or $V_a V_b^*$:

Lemma 4 (Split and identity, general form). *Let $U, K, A, H \in \mathbb{F}[X]_{<N}$ and $v \in \mathbb{F}$. (1) There are $A, H \in \mathbb{F}[X]_{<N}$ with $XUK = A + vX^N + X^{N+1}H$ iff $v = [X^{N-1}]UK$. (2) $\Phi = XUK - A - vX^N - X^{N+1}H$ has degree at most $2N$, and $\Phi = 0$ implies $v = [X^{N-1}]UK$.*

Proof. As in KF Lemmas 4.10 and 4.12: $\deg(UK) \leq 2N-2$, and the three parts of the right-hand side occupy the disjoint degree ranges $[0, N-1]$, $\{N\}$, $[N+1, 2N]$; compare the coefficients of X^N . $\square$

Lemma 5 (Reversed words). *For $w \in \mathbb{F}^L$ put $w^*(\xi) = \xi^{N-1}w(\xi^{-1})$. Then: (1) $w = \text{ev}_L(P)$ with $P \in \mathbb{F}[X]_{<N}$ implies $w^* = \text{ev}_L(P^*)$; in particular $w \mapsto w^*$ maps $\mathcal{C}_0$ onto $\mathcal{C}_0$; (2) $w \mapsto w^*$ is a linear bijection of $\mathbb{F}^L$, and $w^* = c^*$ at ξ iff $w = c$ at ξ^{-1} ; (3) a set $T \subseteq L$ is fibre-closed iff $T^{-1} = \{\xi^{-1} : \xi \in T\}$ is, and $|T^{-1}| = |T|$; (4) $h(\xi) = w^*(\xi)$ is computed from the value of w at $\xi^{-1} \in L$, and a fold check at $\pm\xi$ reads w on the fibre $\{\pm\xi^{-1}\}$.*

Proof. L is a group, so $\xi \mapsto \xi^{-1}$ is a bijection of L mapping $-\xi$ to $-\xi^{-1}$ (as $-1 \in L$); and $\xi^{N-1} \neq 0$. (1) $\xi^{N-1}P(\xi^{-1}) = P^*(\xi)$, and $P \mapsto P^*$ is a bijection of $\mathbb{F}[X]_{<N}$. (2)–(4) follow from the first sentence. $\square$

2 The inner-product protocol

Definition 6 (Π_{IP}). Instance: oracles $w_a, w_b \in \mathbb{F}^L$ and $S \in \mathbb{F}$; honestly $w_a = \text{Enc}^T(a)$, $w_b = \text{Enc}^T(b)$ and $S = \sum_w a(w)b(w)$.

1. The prover sends $w_A = \text{ev}_L(A)$, where $XV_a V_b^* = A + SX^N + X^{N+1}H$ (Lemma 4); the verifier sends $\beta \in \mathbb{F}$.
2. Put the virtual words w_b^* (Lemma 5) and $h(\xi) = (\xi w_a(\xi) w_b^*(\xi) - w_A(\xi) - S\xi^N)/\xi^{N+1}$, and

$$w_0 = w_a + \beta w_b^* + \beta^2 w_A + \beta^3 h.$$

3. Run $\Pi_{\text{FT}}[\ell, \kappa]$ (or Π_{FT}^C , KF Prop. 5.28) on w_0 .

Per query point the verifier reads $w_a(\pm\xi)$, $w_b(\pm\xi^{-1})$ and $w_A(\pm\xi)$, and does $O(1)$ extra field operations beyond KF (no kernel to evaluate).

Theorem 7 (Completeness). *If $w_a = \text{ev}_L(V_a)$, $w_b = \text{ev}_L(V_b)$ and $S = \sum_w a(w)b(w)$, the honest prover is accepted with probability 1.*

Proof. By Lemma 3 and Lemma 4(1), $A, H \in \mathbb{F}[X]_{<N}$ exist; by Lemma 5(1) $w_b^* = \text{ev}_L(V_b^*)$; evaluating the identity at ξ gives $h = \text{ev}_L(H)$. So $w_0 = \text{ev}_L(V_a + \beta V_b^* + \beta^2 A + \beta^3 H)$ with a polynomial of degree $< N$, and KF Lemma 5.9(1) applies. $\square$

Definition 8 (Relation). $\mathcal{R}_{\text{IP}}^\delta$: $((w_a, w_b; S), (a, b))$ with $\Delta_0^{\text{fib}}(w_a, \text{Enc}^T(a)) \leq \delta$, $\Delta_0^{\text{fib}}(w_b, \text{Enc}^T(b)) \leq \delta$ and $\sum_w a(w)b(w) = S$.

Lemma 9 (Batching, degree 3). *Let $\mathcal{G}$ be the set of β for which w_0 agrees with some $c \in \mathcal{C}_0$ on a fibre-closed $T \subseteq L$ with $|T| \geq (1 - \delta)M$. If $|\mathcal{G}| > 3M$, then $(w_a, w_b; S)$ has a witness in $\mathcal{R}_{\text{IP}}^\delta$.*

Proof. The words w_0 form the curve of degree 3 generated by (w_a, w_b^*, w_A, h) , and every $\beta \in \mathcal{G}$ has $\Delta(w_0, \mathcal{C}_0) \leq \delta$. By correlated agreement for curves (KF Thm 2.16, from [2], with $t = 3$) there are $P_0, P_1, P_2, P_3 \in \mathbb{F}[X]_{<N}$ and a set Agr, $|\text{Agr}| \geq (1 - \delta)M$, on which $w_a = \text{ev}(P_0)$, $w_b^* = \text{ev}(P_1)$, $w_A = \text{ev}(P_2)$, $h = \text{ev}(P_3)$ (take Agr maximal).

Value. Let $\Phi = XP_0P_1 - P_2 - SX^N - X^{N+1}P_3$. On Agr, $\Phi(\xi) = \xi w_a(\xi)w_b^*(\xi) - w_A(\xi) - S\xi^N - \xi^{N+1}h(\xi) = 0$ by the definition of h . Since $\deg \Phi \leq 2N < (1 - \delta)M \leq |\text{Agr}|$ (KF Lemma 7.2, which uses $R \geq 2$), $\Phi = 0$, and Lemma 4(2) gives $S = [X^{N-1}]P_0P_1$. Let a, b be the tables with $V_a = P_0$ and $V_b = P_1^*$; then $V_b^* = P_1$ and $S = \sum_w a(w)b(w)$ by Lemma 3.

Bad challenges. For $\xi \notin \text{Agr}$, $d_\xi(Z) = \sum_{i=0}^3 Z^i(u_i - \text{ev}(P_i))(\xi)$, with $(u_0, \dots, u_3) = (w_a, w_b^*, w_A, h)$, is a nonzero polynomial of degree ≤ 3 ; its roots over all such ξ form a set Bad with $|\text{Bad}| \leq 3\delta M < 3M < |\mathcal{G}|$. Take $\beta \in \mathcal{G} \setminus \text{Bad}$ with its c and T , and $\hat{c} = \sum_i \beta^i \text{ev}(P_i) \in \mathcal{C}_0$. Then c and $\hat{c}$ agree with w_0 on $T \cap \text{Agr}$, of size $\geq (1 - 2\delta)M \geq N$, so $c = \hat{c}$; and for $\xi \in T$, $d_\xi(\beta) = w_0(\xi) - \hat{c}(\xi) = 0$ forces $\xi \in \text{Agr}$. So $T \subseteq \text{Agr}$.

Distances. $w_a = \text{ev}(V_a)$ on the fibre-closed T , so $\Delta_0^{\text{fib}}(w_a, \text{Enc}^T(a)) \leq \delta$ (KF Lemma 5.11(2)). $w_b^* = \text{ev}(V_b^*) = \text{ev}(V_b)^*$ on T , so by Lemma 5(2) $w_b = \text{ev}(V_b)$ on T^{-1} , which is fibre-closed of the same size (Lemma 5(3)); so $\Delta_0^{\text{fib}}(w_b, \text{Enc}^T(b)) \leq \delta$. $\square$

Theorem 10 (Soundness). *If $(w_a, w_b; S)$ has no witness in $\mathcal{R}_{\text{IP}}^\delta$, every prover is accepted with probability at most*

$$\frac{3M}{|\mathbb{F}|} + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa < \frac{4M}{|\mathbb{F}|} + (1 - \delta)^\kappa.$$

Proof. Identical to KF Thm 7.8, with Lemma 9 in place of KF Lemma 7.4: for $\beta \notin \mathcal{G}$ (probability $\geq 1 - 3M/|\mathbb{F}|$) and good r (probability $\geq 1 - \varepsilon_{\text{fold}}$, KF Thm 5.24(1)), the codeword chain (KF Prop. 5.23) shows that $\text{dens}_\ell(w_\ell) \geq 1 - \delta$ would put β in $\mathcal{G}$, the witness set $\mathcal{W}_\ell(w_\ell)$ being fibre-closed (KF Lemma 7.7); so the query term is at most $(1 - \delta)^\kappa$ (KF Lemma 3.3). $\square$

Round-by-round knowledge soundness, the extractor (decode w_a and w_b at radius δ^*) and binding follow as in KF Thm 7.11 and Cor. 7.12, with round-1 error $3M/|\mathbb{F}|$.

Remark 11 (One commitment, three kinds of openings). The same table-form commitment $\text{Enc}^T(f)$ supports: evaluation openings at any z (KF's protocol with the kernel E_z^* of Lemma 2, evaluated by the verifier with $O(n)$ operations); hypercube sums ($z = (\frac{1}{2}, \dots, \frac{1}{2})$ up to 2^n , or the kernel $\prod_k (1 + X^{2^{k-1}})$ directly); and inner products with any other committed table (Π_{IP}). Inner products with a *public* table p are the special case $K = V_p^*$, useful when $V_p^*(\xi)$ is cheap to evaluate (product form). All of these batch with KF's batched openings (KF Thm 11.2).

3 Towards a sumcheck-free Hadamard check

SNARKs of Spartan/HyperPlonk type need, beyond linear functionals and inner products, the statement $a(w)b(w) = c(w)$ for all w (a *Hadamard check*). The standard reduction takes a random weighting ρ and checks $\sum_w \rho(w)(a(w)b(w) - c(w)) = 0$, a degree-3 sum.

Geometric weights. Take $\rho(w) = \gamma^w$ for a random $\gamma \in \mathbb{F}$. If $d = a \circ b - c \neq 0$, then $\sum_w \gamma^w d(w)$ is a nonzero polynomial in γ of degree $\leq N - 1$, so it vanishes with probability $\leq (N - 1)/|\mathbb{F}|$. With these weights,

$$\sum_w \gamma^w a(w)b(w) = [X^{N-1}] V_a(\gamma X) V_b^*(X), \quad \sum_w \gamma^w c(w) = V_c(\gamma).$$

Both are one extracted coefficient and one evaluation; the obstruction is that $V_a(\gamma\xi)$ and $V_c(\gamma)$ are values of committed polynomials *outside* L .

Proposal. The prover sends an oracle $w_Q = \text{ev}_L(Q)$ with $Q(X) = V_a(\gamma X)$ and two values $y_1 = V_a(\gamma\theta)$, $y_2 = Q(\theta)$ for a verifier point $\theta \in \mathbb{F} \setminus L$, and $y_3 = V_c(\gamma)$. The verifier adds to the batch the DEEP quotients

$$\frac{w_a(\xi) - y_1}{\xi - \gamma\theta}, \quad \frac{w_Q(\xi) - y_2}{\xi - \theta}, \quad \frac{w_c(\xi) - y_3}{\xi - \gamma},$$

checks $y_1 = y_2$, and runs Π_{IP} on $(w_Q, w_b; y_3)$, all in one folding test. Cost over Π_{IP} : one more oracle (w_Q) and three field elements; the verifier's work per query stays $O(1)$ beyond FRI.

Question 12. Prove soundness of the proposal in the unique-decoding regime: if Δ_0^{fib} -closest code-words satisfy $a \circ b \neq c$, the acceptance probability is at most $O(M/|\mathbb{F}|) + (N-1)/|\mathbb{F}| + (1-\delta)^\kappa$. The two steps to establish are (i) the DEEP step: agreement of the quotient words with low-degree polynomials on a large set forces $V_{\hat{a}}(\gamma\theta) = y_1$ and $\hat{Q}(\theta) = y_2$ for the decoded $\hat{a}, \hat{Q}$, and hence $\hat{Q}(X) = V_{\hat{a}}(\gamma X)$ except with probability $\leq N/|\mathbb{F}|$ over θ ; (ii) the combination of (i) with Lemma 9 in a single batching lemma for a curve of degree 6.

Remark 13 (What a positive answer gives). A sumcheck-free Hadamard check with FRI-only machinery, a verifier that is an FRI verifier plus $O(n)$ operations per query, and the same commitments as evaluation openings (Lemma 2). With linear functionals, inner products and Hadamard checks, the constraint system of R1CS-type SNARKs reduces to these three primitives except for the sparse matrix-vector products Az, Bz, Cz , which Spartan handles with its own sparse-polynomial commitment; whether those also admit a coefficient-extraction form is the next question.

4 Relation to known techniques and status

Writing an inner product as the middle coefficient of a product with a reversed polynomial is classical, and appears in pairing-based multilinear commitments (Mercury [3] uses the table form with KZG). DEEP quotients for out-of-domain values are standard in FRI-based STARKs. The new points here are: (1) the reversed virtual word w_b^* , which reads the second vector from the *same* Reed-Solomon commitment thanks to $L^{-1} = L$; (2) the reuse of Kronecker-FRI's opening polynomial, virtual word and single folding test, with a fully proved soundness bound (Lemma 9); (3) the table-form evaluation kernel E_z^* , which gives evaluation openings on the same commitments.

Prover cost and the sumcheck baseline. The prover of Π_{IP} computes one product $V_a V_b^*$ of two polynomials of degree $< N$ over the base field (three NTTs of size $2N$), one NTT of size M and one Merkle tree for w_A , and then runs the folding test; every round-1 object lies in the base field, since V_a, V_b do. The natural baseline is the sumcheck route: n rounds of a degree-2 sumcheck on $\sum_w a(w)b(w)$, followed by openings of a and b at the random point. For that baseline the fastest prover we know of in our own work is the tree circuit of [4], which produces the evaluations, hypercube sums and sumcheck round polynomials of a multilinear polynomial from one binary-tree recursion; the comparison of the two routes at equal soundness is left to the implementation paper.

Status.

- *Lean 4.* Sections 1–2 are formalised in the Kronecker-FRI Lean project, files `TableForm.lean` and `InnerProduct.lean`, without `sorry` and with the single axiom of that project (correlated agreement for curves, KF Thm 2.16, here with $t = 3$). Lemma 2: `eval_table`;

Lemma 3: `inner_product`; Lemma 4: `splitG_iff`, `natDegree_PhiG_le`, `identityG`; Lemma 5: `wrev_ev`, `wrev_eq_iff`, `inv_image_closed`, `ncard_inv_image`; completeness: `honestIP_prob_one`; Lemma 9: `batching_IP`; soundness: `soundness_IP`, and `soundnessC_IP` for any folding arity 2^k ; knowledge and binding: `knowledge_IP`, `binding_IP`. The common step of KF Lemma 7.4 and Lemma 9, correlated agreement on one fibre-closed set for a curve of any degree, is the separate theorem `curve_agreement`.

- *Rust*. The module `ip` of the Kronecker-FRI crate implements Π_{IP} with Merkle caps, arity 2^k and Fiat–Shamir, on the commitments of `pcs`. Its tests check Lemmas 3, 4 and 5(1), completeness for $n \leq 10$, two rates and arities 2 to 16, and the rejection of three cheating provers (a wrong S ; a wrong S absorbed into a polynomial A of degree N , as in KF Remark 4.13; opening polynomials computed from a table $b' \neq b$).
- Section 3 is a proposal; Question 12 is open.

n	Π_{KF} (one evaluation)			Π_{IP} (one inner product)		
	prove (ms)	verify (ms)	proof (KiB)	prove (ms)	verify (ms)	proof (KiB)
12	8.4	1.95	138.1	6.8	1.95	171.4
14	33.6	2.44	182.3	27.1	2.45	233.4
16	141.0	3.04	249.8	112.8	3.10	308.1
18	556.7	3.72	325.0	451.6	3.74	395.3
20	2692.3	4.35	391.7	1996.2	4.41	471.1

Table 1: Configuration KF-8 of the Kronecker-FRI paper (rate 1/4, arity 8, caps of 128 nodes, $\ell = n - 8$), 148 queries, 32-byte salts, $\mathbb{F}_{p^2}$ over Goldilocks; one core of an Intel Xeon at 2.10 GHz; medians; commitments not included. Source: `bench/ip.csv`, `cargo run --release --example ip.bench`.

An inner product of two committed tables costs less prover time than one evaluation opening (the round-1 work is in the base field), the same verifier time, and about 20% more proof, the extra openings of w_b on the inverse cosets.

Next. Question 12 (DEEP consistency and the batching lemma for a curve of degree 6), then the batching of Π_{IP} with evaluation openings (KF Thm 11.2) in one folding test.

References

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